Aviation ERP – Full System Report

Comprehensive system documentation and source appendix

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Workspace: D:\aviation_erp

Executive Summary

This report aggregates project documentation, architecture notes, assets, and the full source code for the Aviation ERP system into a single long-form PDF. It includes screenshots, database schemas, migrations, backend services, frontend application code, serverless functions, and supporting scripts.

Project Overview

```
# Student ERP

<p align="center">
  <b>Aviation training management system for small institutes</b><br/>
  Attendance, fees, role-based access, and Excel sync in one app.
</p>

<p align="center">
  <img alt="Backend" src="https://img.shields.io/badge/Backend-FastAPI-0f766e">
  <img alt="Database" src="https://img.shields.io/badge/Database-SQLite-1d4ed8">
  <img alt="Frontend" src="https://img.shields.io/badge/Frontend-HTML%2FCSS%2FJS-1f2937">
  <img alt="Status" src="https://img.shields.io/badge/Status-Live%20on%20Cloudflare-16a34a">
</p>

## Live URL

- https://student-erp.pages.dev

## What Is This?

Student ERP is a local-first web app for managing students in an aviation training workflow.

It helps you:
- track attendance daily,
- record fee payments with receipts,
- separate superuser and student views,
- sync attendance from an Excel source file.

## Who Is It For?

- Training institutes that want a lightweight internal tool.
- Teams that currently maintain attendance in Excel but need a web dashboard.

## Main Features

- Role-based login:
  - Students can see only their own data.
- Attendance:
  - Mark attendance from the web UI.
  - Date-wise attendance view.
  - Sync attendance from `attendance_master.xlsm`.
- Fees:
  - Record payments per student.
  - Upload receipt files (`png`, `pdf`, and other file types).
  - Student-facing remaining balance view.
  - Student Razorpay payment flow with post-payment invoice PDF download.
  - Staff can generate invoice PDF from fee entries in portal.
- Session handling:
  - Session expires after 5 minutes of inactivity.
  - Refresh keeps your current tab while session is valid.

## Quick Start

### 1. Install dependencies

```bash
pip install -r requirements.txt
```

For development and tests:

```bash
pip install -r requirements-dev.txt
```
2. Start backend
```

```bash
uvicorn backend.main:app --reload
```

3. Open frontend

Open:
- `frontend/index.html`

Backend URL used by frontend:
- `http://127.0.0.1:8000`

Default Login (Local Demo)

- Students:
  - Username: student ID
  - Password: generated in `passwords.txt` (local file, gitignored)

Project Structure

```text
aviation_erp/
|- backend/
|  |- main.py
|  |- db.py
|  |- schema.sql
|  |- import_attendance_excel.py
|  |- erp.db              # local runtime DB (gitignored)
|  `- receipts/           # uploaded receipts (gitignored)
|- frontend/
|  |- index.html
|  |- style.css
|  `- app.js
|- attendance_master.xlsm
|- passwords.txt          # generated local credentials (gitignored)
|- .gitignore
`- README.md
```

API Snapshot

- `POST /login`
- `GET /auth/me`
- `GET /students`
- `POST /students`
- `GET /students/{student_id}/balance`
- `GET /students/{student_id}/attendance`
- `GET /students/{student_id}/fees`
- `GET /attendance/recent`
- `GET /attendance/by-date`
- `POST /attendance/record`
- `POST /attendance/sync`
- `POST /fees/record`
- `GET /fees/recent`
- `GET /fees/{fee_id}/invoice`
- `GET /payments/gateway-status`
- `POST /payments/razorpay/order`
- `POST /payments/razorpay/verify`
- `GET /reports/summary`

Tests

Run backend tests with:

```bash
pytest -q
```

Attendance and Excel Behavior

- Attendance recorded in UI is saved to:
  - `backend/erp.db`
  - `attendance_master.xlsm`
- Excel format used:
  - date: `dd-mm-yyyy`
  - status: `P` or `A`
- `Sync From Excel` reloads attendance from Excel into the DB.

Screenshots

Login
![Login](docs/screenshots/login.png)

Superuser Dashboard
![Superuser Dashboard](docs/screenshots/superuser-dashboard.png)

Attendance Tab
![Attendance Tab](docs/screenshots/attendance-tab.png)

Student View
![Student View](docs/screenshots/student-view.png)

Production Notes

Current auth/session setup is suitable for local/internal usage. For production:
- hash passwords (`bcrypt`),
- use a stronger session strategy (JWT or persistent session store),
- tighten CORS policy,
- enforce HTTPS and secure cookie handling.

Cloudflare Pages Deployment (Free Tier)

This repo now includes:
- Pages Functions API: `functions/api/[[path]].js`
- D1 migration: `migrations/0001_init.sql`
- Wrangler config: `wrangler.toml`

1. Install and login

```bash
npm install -g wrangler
wrangler login
```

2. Create D1 and R2

```bash
wrangler d1 create student-erp-db
wrangler r2 bucket create student-erp-files
```

Update `wrangler.toml`:
- set `database_id` from D1 create output
- keep `binding = "DB"` and `binding = "ERP_FILES"`

3. Apply schema migration

```bash
wrangler d1 migrations apply student-erp-db
```

4. Set Razorpay secrets (optional but required for payments)

```bash
wrangler pages secret put RAZORPAY_KEY_ID --project-name student-erp
wrangler pages secret put RAZORPAY_KEY_SECRET --project-name student-erp
```

Important:
- secret names must be exactly `RAZORPAY_KEY_ID` and `RAZORPAY_KEY_SECRET`
- add them to the **Production** environment in Pages
- redeploy after updating secrets

5. Deploy

```bash
wrangler pages deploy frontend --project-name student-erp
```

After deploy:
- frontend is served from Pages
- API available at `/api/*` via Functions
- storage uses D1 + R2
- student payments show `Razorpay ready` only when both secrets are configured
Local Razorpay setup (developer machine)

- For FastAPI local run, create `backend/.env`:

```bash
RAZORPAY_KEY_ID=...
RAZORPAY_KEY_SECRET=...
```

- For `wrangler pages dev`, create `.dev.vars`:

```bash
RAZORPAY_KEY_ID=...
RAZORPAY_KEY_SECRET=...
SESSION_TIMEOUT_SECONDS=300
```

Both files are local-only and gitignored.

Attendance import on Cloudflare

- In Attendance tab, use `Sync From Excel` with a file selected.
- Parser currently supports `CSV` for automatic import.
- `.xlsx` / `.xlsm` files are uploaded to R2 but not parsed automatically.
- Recommended flow:
  1. Export `attendance_master.xlsm` sheet to CSV
  2. Upload CSV in the Attendance section
  3. Click `Sync From Excel`

Architecture Notes

Architecture and Endpoint Mapping

Overview
The Student ERP system uses a dual-backend strategy:
1. **Local Backend**: FastAPI-based (`backend/main.py`), using SQLite (`backend/erp.db`).
2. **Production Backend**: Cloudflare Pages Functions (`functions/api/[[path]].js`), using Cloudflare D1 and R2.

The frontend is a single-page application (SPA) that can communicate with either backend depending on the host.

Endpoint Mapping Discrepancies

Feature	FastAPI (`backend/main.py`)	Cloudflare (`functions/api/[[path]].js`)	Status
Auth/Login	Yes	Yes	Equivalent
Student Management	Yes	Yes	Equivalent
Attendance	Yes	Yes	Equivalent
Fees	Yes	Yes	Equivalent
Profile	Yes	Yes	Equivalent
Timetable	Yes	Yes	Equivalent
Interviews	Yes	Yes	Equivalent
Announcements	Yes	Yes	Equivalent
Notifications	Yes	Yes	Equivalent
Admissions	Yes	Yes + Photo/PDF Replace	Discrepancy (Photo/PDF)
Activity Logs	Yes	Yes	Equivalent
Tests	Yes	Yes	Equivalent
Leads	Yes	Yes	Equivalent
Chatbot	Yes	Yes	Equivalent
Parent Portal	Yes	Yes	Equivalent

Frontend Modularization
The frontend was originally in a single `frontend/app.js` file. It has been partially or fully migrated to modular scripts in `frontend/js/`.

- `frontend/index.html` currently loads `/js/main.js` as a module.
- `frontend/js/main.js` imports all other modules and attaches them to the `window` object to maintain compatibility with inline event handlers in `index.html`.

Frontend Modularization Status
The frontend has been fully refactored from legacy monolithic files (`frontend/app.js` and `frontend/js/app-core.js`) into a granular modular structure within `frontend/js/`.

- **Module Structure**: Business logic is decoupled into ~20 specific modules (e.g., `auth.js`, `attendance.js`, `pdf.js`, `chatbot.js`).
- **Global Compatibility**: `frontend/js/main.js` serves as the entry point, aggregating all modules and exposing necessary functions to the `window` object to maintain compatibility with existing inline HTML event handlers.
- **Loading Mechanism**: `frontend/index.html` loads the application via an ES module import:
  ```html
  <script type="module">
    import(`/js/main.js?v=${Date.now()}`)
      .then(() => { window.__APP_READY__ = true; })
  ```

Outcome: Monolithic legacy files have been removed. The modular architecture improves maintainability while preserving full feature parity.

Repository File Tree

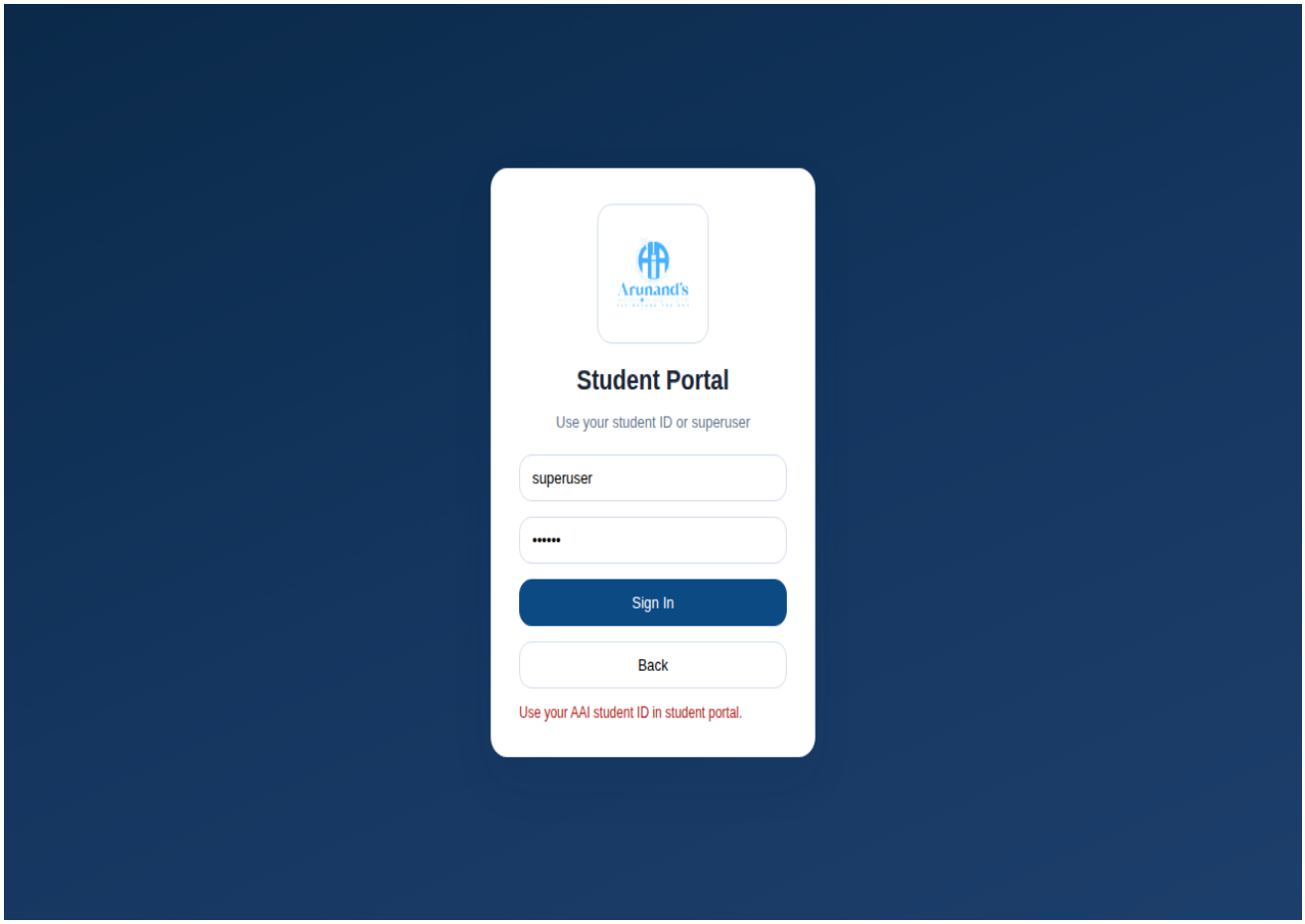
```
- .gitattributes
- .gitignore
- architecture_mapping.md
- attendance_master.csv
- attendance_master.xlsm
- backend
  - backend/__init__.py
  - backend/db.py
  - backend/import_attendance_excel.py
  - backend/main.py
  - backend/schema.sql
- debug_app_failed.png
- docs
  - docs/ERP_Full_Report.pdf
  - docs/fee_receipt_sample.pdf
  - docs/screenshots
    - docs/screenshots/attendance-tab.png
    - docs/screenshots/attendance-tab.svg
    - docs/screenshots/login.png
    - docs/screenshots/login.svg
    - docs/screenshots/student-view.png
    - docs/screenshots/student-view.svg
    - docs/screenshots/superuser-dashboard.png
    - docs/screenshots/superuser-dashboard.svg
- final_state.png
- frontend
  - frontend/assets
    - frontend/assets/favicon.svg
    - frontend/assets/logo.pdf
    - frontend/assets/logo.png
  - frontend/index.html
  - frontend/js
    - frontend/js/activity.js
    - frontend/js/admissions.js
    - frontend/js/api-client.js
    - frontend/js/attendance.js
    - frontend/js/auth.js
    - frontend/js/chatbot.js
    - frontend/js/config.js
    - frontend/js/content.js
    - frontend/js/errors.js
    - frontend/js/fees.js
    - frontend/js/leads.js
    - frontend/js/main.js
    - frontend/js/parent.js
    - frontend/js/pdf.js
    - frontend/js/reports.js
    - frontend/js/state.js
    - frontend/js/students.js
    - frontend/js/tests.js
    - frontend/js/ui.js
    - frontend/js/utils.js
  - frontend/manifest.json
  - frontend/service-worker.js
  - frontend/style.css
- functions
  - functions/api
    - functions/api/[[path]].js
- Internship_CIA_PPT_updated.pptx
- migrations
  - migrations/0001_init.sql
  - migrations/0002_admissions.sql
  - migrations/0003_activity_log.sql
  - migrations/0004_fee_policies.sql
  - migrations/0005_student_profiles.sql
- README.md
- requirements-dev.txt
- requirements.txt
- run_server.py
- sample_fee_receipt.pdf
- scripts
  - scripts/generate_erp_report.py
  - scripts/seed_d1_from_excel.sql
  - scripts/set_remaining_fee_150k.sql
```

```
- staff_dashboard.png
- student_dashboard.png
- sync-github.ps1
- tests
  - tests/test_auth_roles.py
- wrangler.toml
```

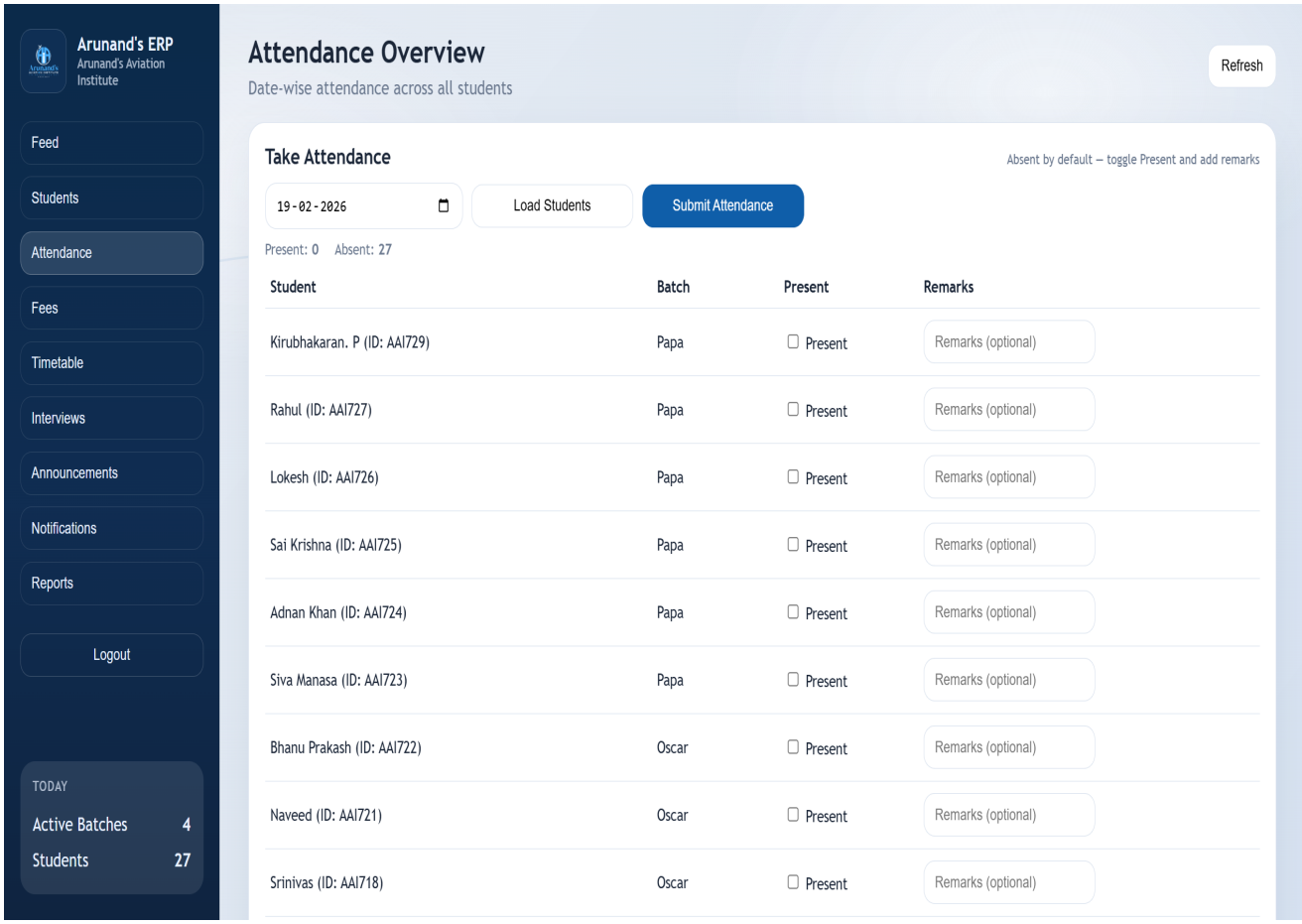
Media Assets

Screenshots and visuals included below.

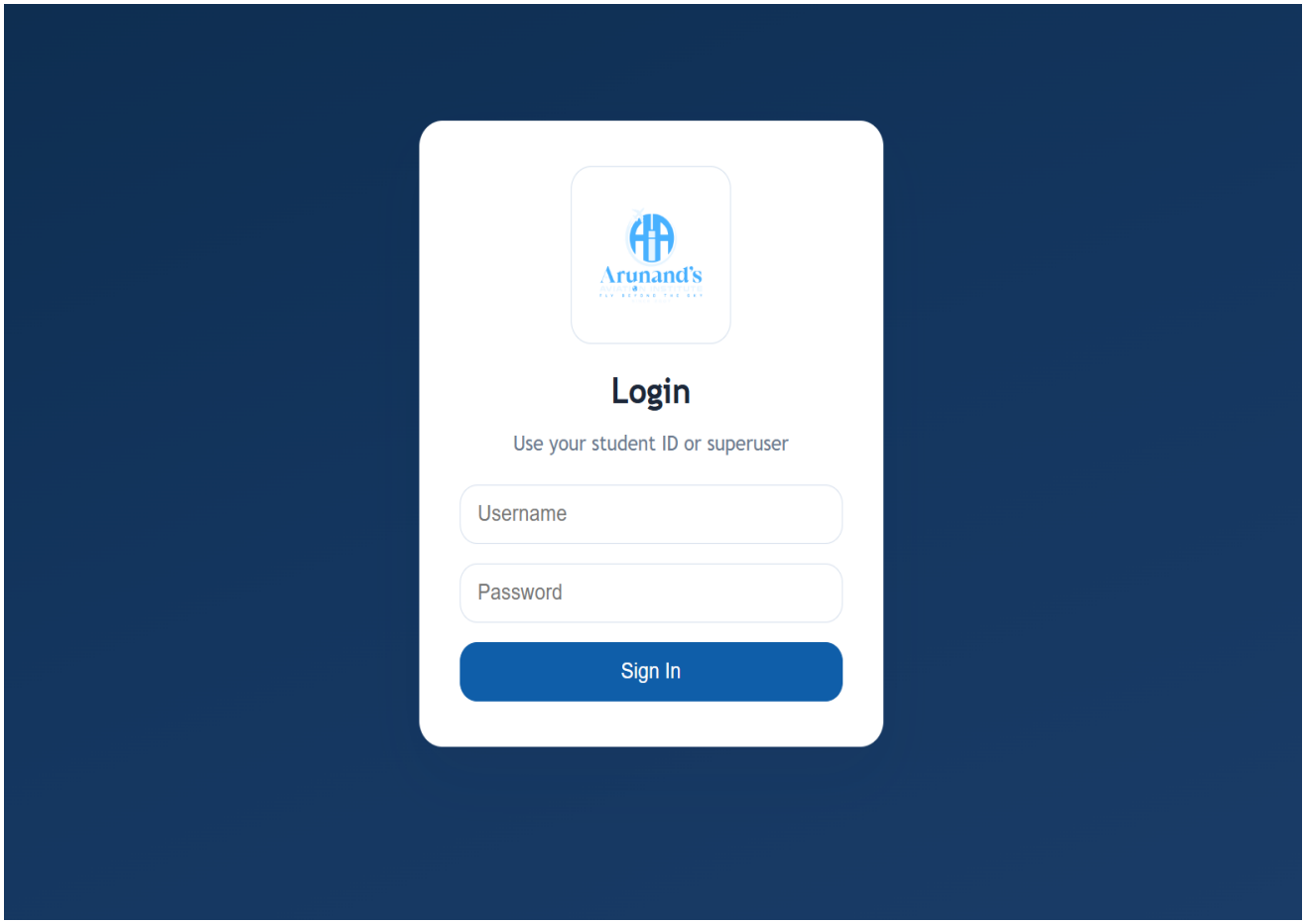
debug_app_failed.png



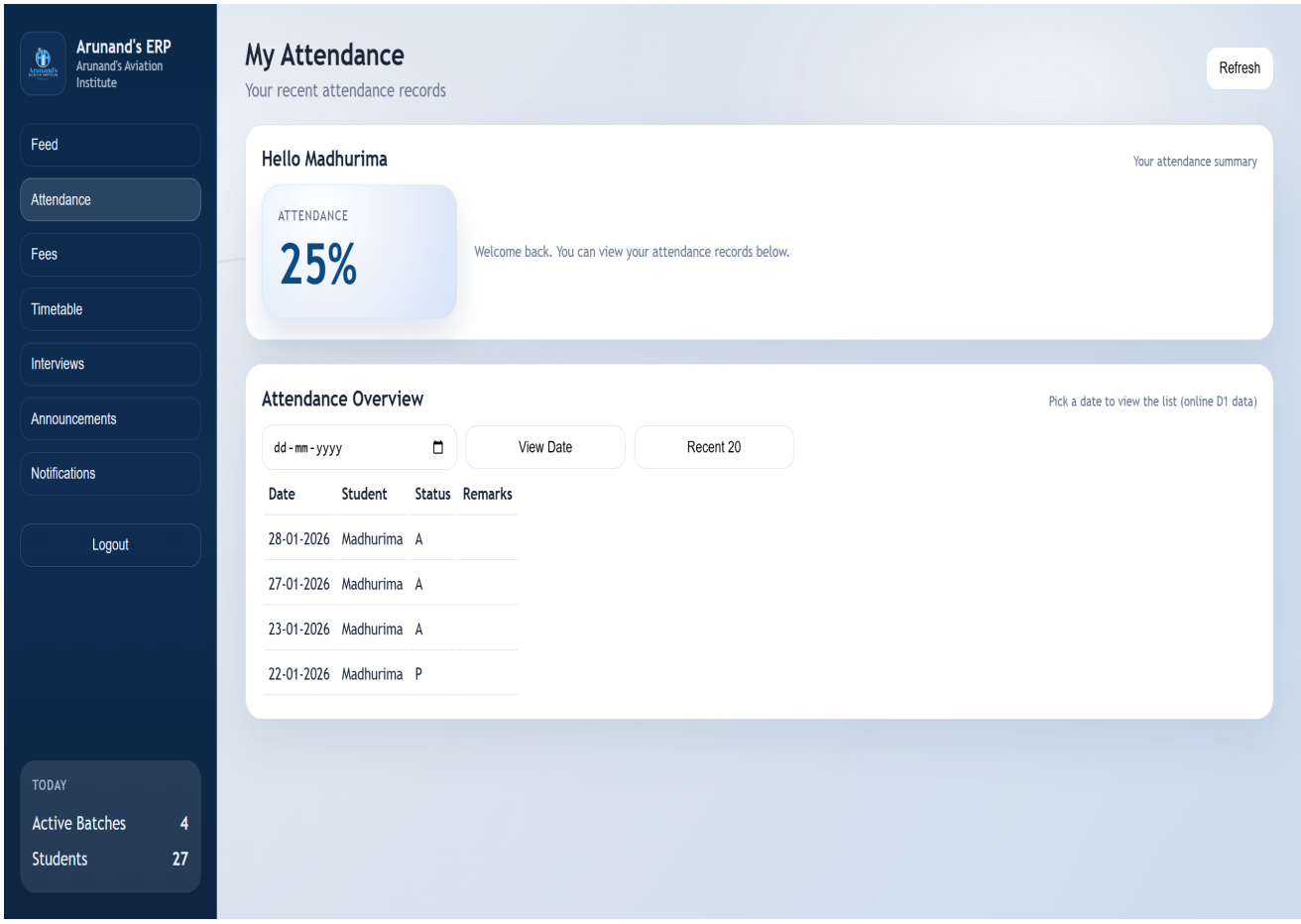
docs/screenshots/attendance-tab.png



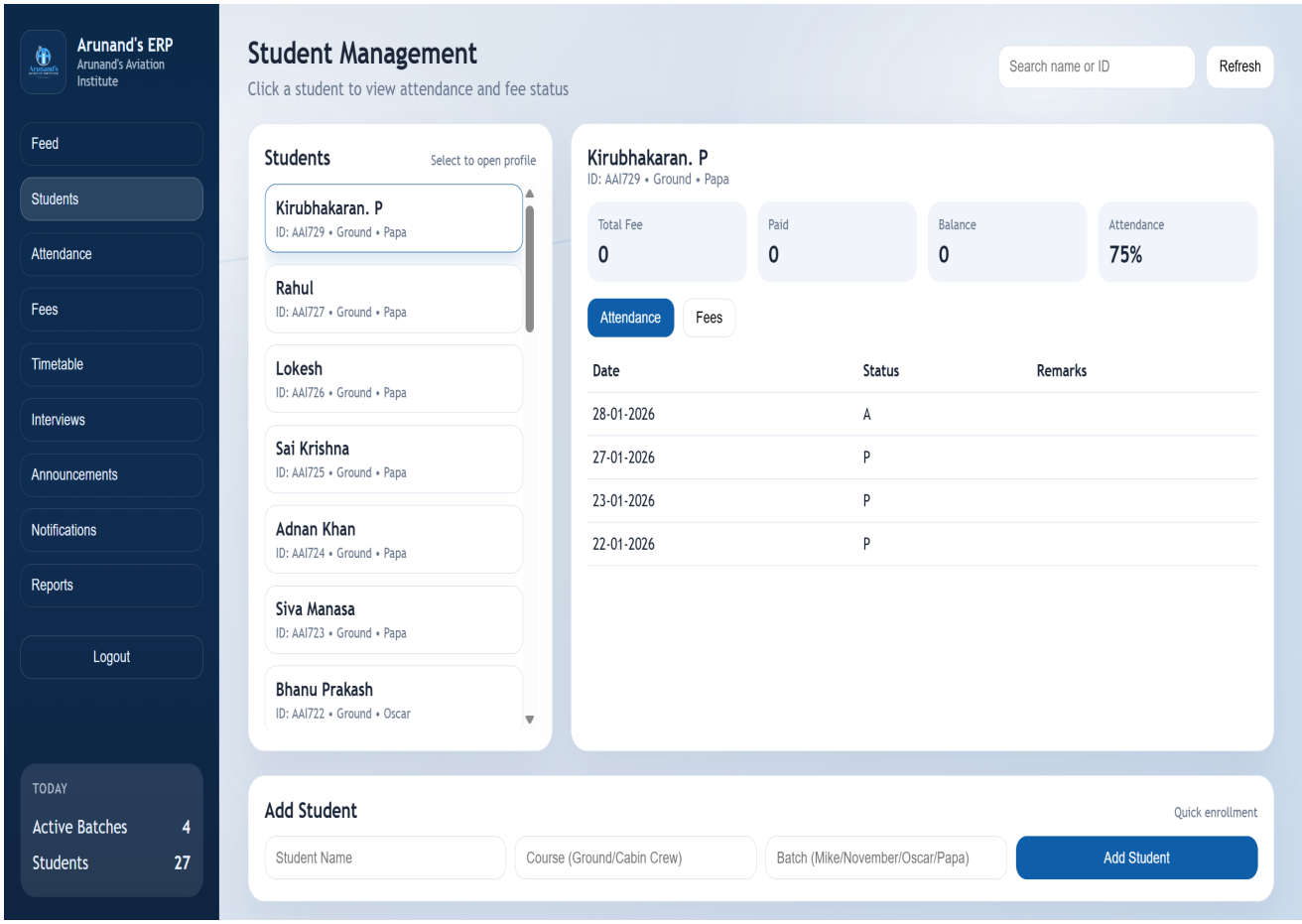
docs/screenshots/login.png



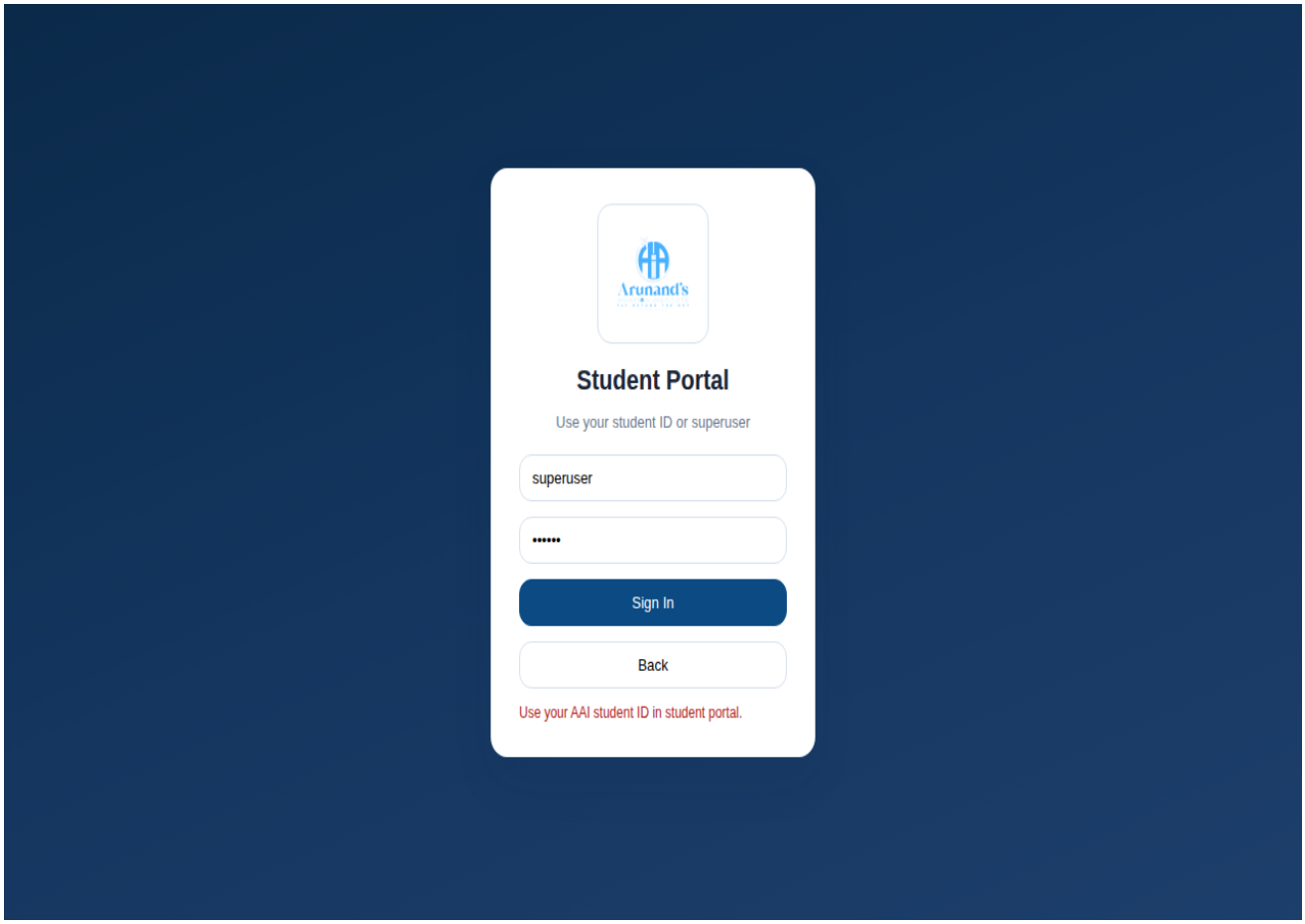
docs/screenshots/student-view.png



docs/screenshots/superuser-dashboard.png

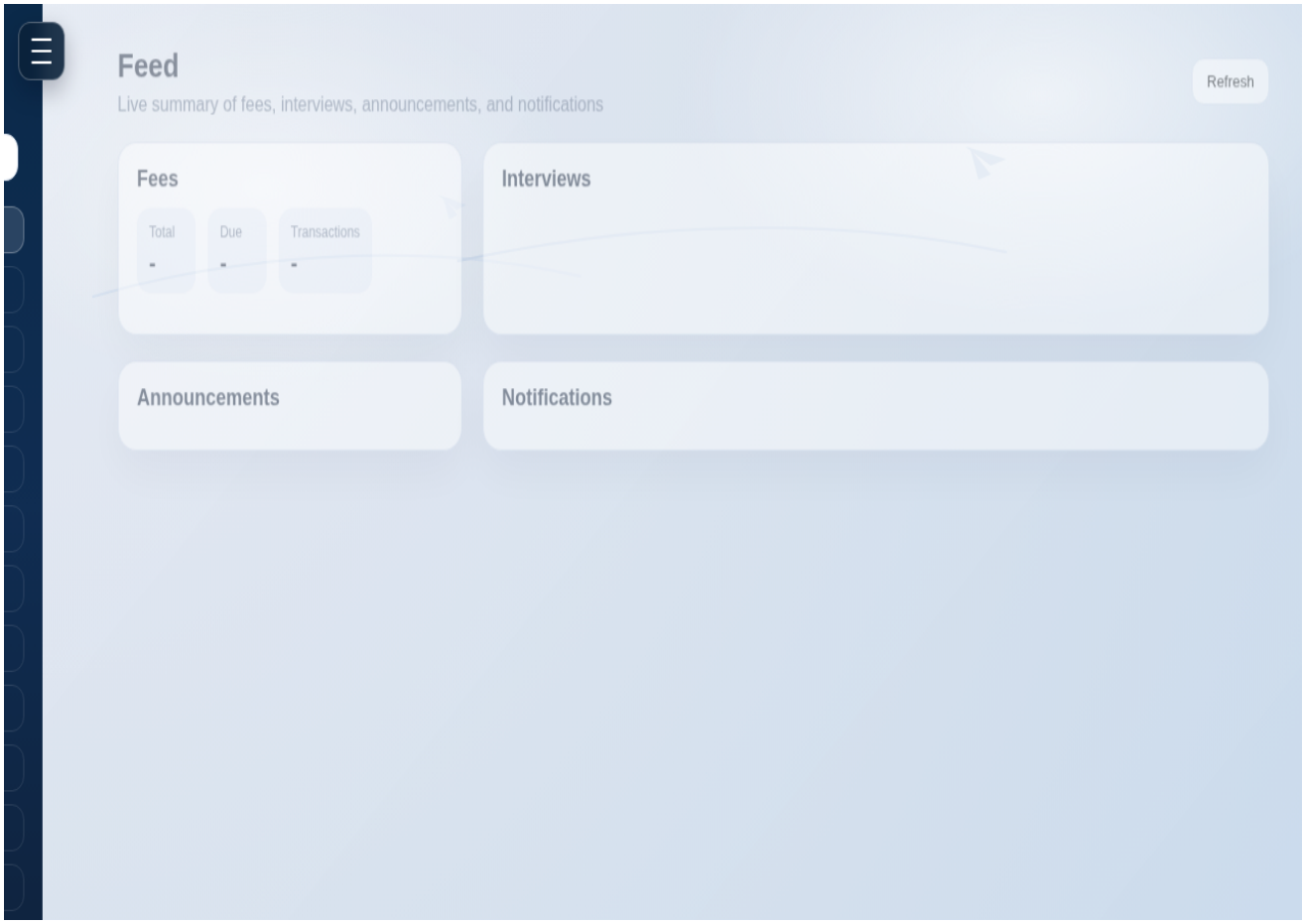


final_state.png



frontend/assets/logo.png

staff_dashboard.png



student_dashboard.png

Feed
Live summary of fees, interviews, announcements, and notifications
Refresh
Fees
Total
Due
Transactions
Interviews
Announcements
Notifications

PDF Assets (Referenced)

docs/ERP_Full_Report.pdf – 3250.2 KB

docs/fee_receipt_sample.pdf – 316.6 KB

frontend/assets/logo.pdf – 224.4 KB

sample_fee_receipt.pdf – 4.9 KB

Source Code Appendix (No Code)

This appendix lists all text-based source files without including their contents, per the requested no-code policy. Sensitive or generated artifacts remain excluded.

attendance_master.csv – 13.0 KB

Contents intentionally omitted (no code included in this report).

backend/__init__.py – 0.0 KB

Contents intentionally omitted (no code included in this report).

backend/db.py – 0.9 KB

Contents intentionally omitted (no code included in this report).

backend/import_attendance_excel.py – 3.9 KB

Contents intentionally omitted (no code included in this report).

backend/main.py – 114.0 KB

Contents intentionally omitted (no code included in this report).

backend/schema.sql – 3.8 KB

Contents intentionally omitted (no code included in this report).

docs/screenshots/attendance-tab.svg – 1.6 KB

Contents intentionally omitted (no code included in this report).

docs/screenshots/login.svg – 1.2 KB

Contents intentionally omitted (no code included in this report).

docs/screenshots/student-view.svg – 2.1 KB

Contents intentionally omitted (no code included in this report).

docs/screenshots/superuser-dashboard.svg – 2.2 KB

Contents intentionally omitted (no code included in this report).

frontend/assets/favicon.svg – 0.3 KB

Contents intentionally omitted (no code included in this report).

frontend/index.html – 57.3 KB

Contents intentionally omitted (no code included in this report).

frontend/js/activity.js – 2.9 KB

Contents intentionally omitted (no code included in this report).

frontend/js/admissions.js – 12.9 KB

Contents intentionally omitted (no code included in this report).

frontend/js/api-client.js – 1.0 KB

Contents intentionally omitted (no code included in this report).

frontend/js/attendance.js – 18.2 KB

Contents intentionally omitted (no code included in this report).

frontend/js/auth.js – 6.8 KB

Contents intentionally omitted (no code included in this report).

frontend/js/chatbot.js – 8.3 KB

Contents intentionally omitted (no code included in this report).

frontend/js/config.js – 2.2 KB

Contents intentionally omitted (no code included in this report).

frontend/js/content.js – 7.3 KB

Contents intentionally omitted (no code included in this report).

frontend/js/errors.js – 0.3 KB

Contents intentionally omitted (no code included in this report).

frontend/js/fees.js – 16.5 KB

Contents intentionally omitted (no code included in this report).

frontend/js/leads.js – 6.9 KB

Contents intentionally omitted (no code included in this report).

frontend/js/main.js – 2.0 KB

Contents intentionally omitted (no code included in this report).

frontend/js/parent.js – 2.9 KB

Contents intentionally omitted (no code included in this report).

frontend/js/pdf.js – 28.6 KB

Contents intentionally omitted (no code included in this report).

frontend/js/reports.js – 1.6 KB

Contents intentionally omitted (no code included in this report).

frontend/js/state.js – 1.0 KB

Contents intentionally omitted (no code included in this report).

frontend/js/students.js – 13.8 KB

Contents intentionally omitted (no code included in this report).

frontend/js/tests.js – 14.5 KB

Contents intentionally omitted (no code included in this report).

frontend/js/ui.js – 12.9 KB

Contents intentionally omitted (no code included in this report).

frontend/js/utils.js – 3.4 KB

Contents intentionally omitted (no code included in this report).

frontend/manifest.json – 0.3 KB

Contents intentionally omitted (no code included in this report).

frontend/service-worker.js – 1.7 KB

Contents intentionally omitted (no code included in this report).

frontend/style.css – 20.8 KB

Contents intentionally omitted (no code included in this report).

functions/api/[[path]].js – 128.5 KB

Contents intentionally omitted (no code included in this report).

migrations/0001_init.sql – 2.9 KB

Contents intentionally omitted (no code included in this report).

migrations/0002_admissions.sql – 0.3 KB

Contents intentionally omitted (no code included in this report).

migrations/0003_activity_log.sql – 0.3 KB

Contents intentionally omitted (no code included in this report).

migrations/0004_fee_policies.sql – 0.2 KB

Contents intentionally omitted (no code included in this report).

migrations/0005_student_profiles.sql – 1.1 KB

Contents intentionally omitted (no code included in this report).

requirements-dev.txt – 0.0 KB

Contents intentionally omitted (no code included in this report).

requirements.txt – 0.1 KB

Contents intentionally omitted (no code included in this report).

run_server.py – 0.1 KB

Contents intentionally omitted (no code included in this report).

scripts/generate_erp_report.py – 8.2 KB

Contents intentionally omitted (no code included in this report).

scripts/seed_d1_from_excel.sql – 32.4 KB

Contents intentionally omitted (no code included in this report).

scripts/set_remaining_fee_150k.sql – 1.0 KB

Contents intentionally omitted (no code included in this report).

sync-github.ps1 – 1.0 KB

Contents intentionally omitted (no code included in this report).

tests/test_auth_roles.py – 2.6 KB

Contents intentionally omitted (no code included in this report).

wrangler.toml – 0.3 KB

Contents intentionally omitted (no code included in this report).

Excluded Sensitive/Generated Files

For safety, the following files or directories were excluded from the report:

.dev.vars, .env, .git, .wrangler, __pycache__, erp.db, node_modules, passwords.txt